

claude-windows / c6028e72-ebfd-4218-89a8-c4d7dd23be96

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c6028e72-ebfd-4218-89a8-c4d7dd23be96\subagents\workflows\wf_49aab888-e10\agent-ab20b50323a40d692.jsonl`  
- SHA-256: `db6f8cd7c5e9510d1269576a29a9a20ca047e3caeb458ed41bb5769abe36f982`  
- Source modified: `2026-06-08T13:13:45+00:00`  
- Imported at: `2026-07-05T16:48:23+00:00`  
- project: `wf_49aab888-e10`  
- session_id: `c6028e72-ebfd-4218-89a8-c4d7dd23be96`
```

Transcript

1. user (2026-06-08T13:04:39.576Z)

DIMENSION = TRAINING COMPUTE PLATFORMS. Use WebSearch + WebFetch. Compare managed + serverless + raw GPU options for training/fine-tuning OUR model: Google Vertex AI (the one the user named), AWS SageMaker, Azure ML, and serverless/independent GPU clouds (Modal, RunPod, Lambda Labs, vast.ai, CoreWeave, Together AI, Paperspace), plus self-hosted/on-prem (the user has one RTX 2070 Super 8GB). For EACH: current GPU pricing (A100/H100/L4/T4, on-demand vs spot), DATA PRIVACY terms (does the provider train on / retain your data? data-residency / India region availability), ease + lock-in, multi-GPU + checkpointing, and fit for a solo Bangalore founder. Privacy is a HARD requirement (our lever is data never being used by third parties). Give concrete \$ where possible and cite URLs.

Keep it aligned to OUR situation:

OUR SITUATION (a local-first racket-sports highlight indexer + rally detector):

- GOAL: an OWNED, on-device-capable model for RALLY TEMPORAL SEGMENTATION (predict rally [start,end] intervals in a match video), evolving toward "Option B" = an OWNED END-TO-END MULTIMODAL model (frames + audio + pose -> rallies/highlights) fine-tuned on our own data. Privacy lever = local inference, user video never leaves the device.
- HARD GUARDRAILS (non-negotiable): (1) base/teacher models for TRAINING must be OPEN Apache-2.0 (or MIT/BSD) ONLY; video-capable bases like Qwen-VL / Gemma family are candidates BUT **Gemma 3 is REJECTED (viral/non-standard license)**. (2) NEVER use paid LLM outputs (Gemini/OpenAI/Anthropic) as TRAINING data ? they are inference/eval/teacher-for-eval only (terms + copyright). (3) Exclude minors from the training pool; biometrics/gait = future (GDPR Art. 9). (4) Generic data schema; coach->athlete is an optional overlay.
- DATA ASSETS WE WANT TO REUSE FOR TRAINING: human-labeled rally [start,end] CSVs per match (currently Adarsh-and-Avi 96 rallies, Kushagra 32, more coming; schema rally_number,start_time,end_time,ending_reason,sport) + per-rally ending_reason labels (winner/forced_error/unforced_error/service_fault/let) + per-video WASB shuttle-trajectory CSVs (Frame,Visibility,X,Y; from a frozen MIT-licensed shuttle detector) + (eval-only, NOT training) Gemini silver labels. Footage = single-camera amateur GoPro, 1080p/4K, 60fps.
- CURRENT PIPELINE: WASB (frozen vision shuttle detector) -> trajectory -> hand-tuned windowing + a net-crossing "gate" + a small distilled LogisticRegression on trajectory features. Honest results: threshold tuning ceilings ~F1 0.73 vs HUMAN labels on one match; a learned model UNDERPERFORMS baseline at n=2 videos. Bottleneck = more golden data + WASB recall, not the method.
- CONSTRAINTS: solo founder, BANGALORE-based (cost-sensitive; cloud GPU pricing + data-residency matter), local hardware = ONE RTX 2070 Super (8GB) + WSL2. User explicitly mentioned Vertex AI but wants MORE options + a detailed comparison.
- THE TASK CLASS is temporal action localization/segmentation (action = "rally"); labels are segment intervals.

Grounding brief:

```
{  
  "owned_model_goal": "Build an OWNED, on-device-capable generative video-QA model (fine-tuned from an Apache-2.0 video VLM base) for RALLY TEMPORAL SEGMENTATION and future end-to-end multimodal sport understanding. Privacy lever: local inference, user video never leaves device."
```

The model plugs into the existing provider registry with paid Gemini as an always-ready fallback. Long-term path (Option B): collapse modular local cues (WASB trajectory, audio hit detection, pose detection) into a single fine-tuned end-to-end model. Near-term (Option A, already shipping): modular signal fusion on frozen or near-frozen bases. The moat is NOT the model weights ? it's the consented badminton-specific video?QA dataset + the eval harness. Owned model becomes primary only after scaffolding (async jobs, multi-tenant, storage/compute abstractions) is rock-solid and the golden-set eval/training corpus is healthy.",

"constraints": [

"Base model MUST be Apache-2.0 clean (commercial-viable, fine-tunable, derivative-wrappable). REJECT Gemma 3 (viral Google PUP), Llama Vision (700M-MAU cap + naming mandate). APPROVED bases: Qwen2.5-VL-7B/32B (Apache-2.0 at those sizes only; 3B/72B custom-licensed), Qwen3-VL-4B/8B, Gemma 4 multimodal (Apache-2.0, 2026), InternVL2.5 (MIT).",

"HARD RULE: Training data must be Apache-2.0 or MIT open-model outputs ONLY. FORBIDDEN: any outputs from paid frontier APIs (Gemini, OpenAI, Anthropic, Claude) as training signal ? they are competing-model clauses and breach terms. Paid APIs are inference/evaluation/teacher-reference only, never training data sources. This is non-negotiable per legal.",

"Exclude minors' video from the training pool BY DEFAULT. Training requires separate, explicit opt-in consent (distinct from processing consent). Per-user training adapters trained only on that user's own consented video; deletion = drop the adapter.",

"Generic data schema (no coach-athlete hardcoding). Coach/athlete overlay is optional, orthogonal. Data model stays reusable for solo users, academies, power users.",

"Training evaluation must use a held-out golden QA eval set (controlled, curated, not user-contributed). Agent-runnable training requires trustworthy pass/fail verdict (not eyeballing). Regression testing locked to golden labels to prevent metric inflation from data leakage.",

"Video-QA fine-tuning works ONLY on labeled data. Currently available: 96 (Adarsh+Avi) + 32 (Kushagra) human-labeled rally [start,end] CSVs per match + Gemini silver labels (eval-only, not training). 5k?20k curated clip?QA pairs is the realistic training band."

],

"data_assets": [

"Human-labeled rally [start,end] CSVs per match (Adarsh-and-Avi 96 rallies, Kushagra 32; schema: rally_number, start_time, end_time, ending_reason, sport). Source: sports-data-collector repo, curate export with commercial-license gate.",

"Per-rally ending_reason labels (winner, forced_error, unforced_error, service_fault, let) ? rally semantics feeding the QA-pair dataset.",

"Per-video WASB shuttle-trajectory CSVs (Frame, Visibility, X, Y) from frozen MIT-licensed WASB-SBDT shuttle detector. Trajectory is a primary local signal for multi-cue fusion.",

"Gemini silver labels (eval and teacher-reference ONLY, explicitly not training-data). Used to measure oracle quality and as a training-time prior (Option A modular fusion), never harvested as outputs to train the owned model.",

"Single-camera amateur GoPro footage: 1080p/4K, 60fps. 22/22 ShuttleSet matches re-acquired at best available resolution (Phase 2 complete). ~26 GB corpus licensed/owned, no user uploads.",

"Golden-set ground-truth expansion via Roora vendor (Bengaluru, initial pilot underway): 3-sport pilot (badminton, pickleball, table tennis) with validated rally CSV schema and annotation guideline.",

"Sibling rallyannotator repo (MIT, open-source, VLC-based): owner can self-produce expert-labeled eval clips. Feed via `import-local` to the collector and `--labels` loaders to the indexer."

],

"task_framing": "***Task: design a training study blueprint for the OWNED video-QA model that clarifies (1) what data is safe/available now, (2) which base model to lock, (3) which fine-tuning backend fits our constraints, and (4) how to stage the transition from heuristics ? Option A modular fusion ? Option B end-to-end owned model.**\\n\\n**Current pipeline (heuristics + frozen detectors):** WASB (frozen vision) ? trajectory ? hand-tuned windowing + net-crossing gate + distilled LogisticRegression on trajectory features. Honest baseline: threshold-tuned F1 ? 0.73 vs human labels (one match); learned model underperforms at n=2 videos. Bottleneck = golden data + WASB recall, not method.\\n\\n**Option A (now ? 6?12 months):** Modular, fully-local cues: WASB trajectory + audio hit-detection (E1 probe AMBER / leaning negative, gated on more golden labels) + pose/court serve-gating (validated but heavy). Each cue swappable, independently testable, generates training signal for the end-to-end model. Fusion happens in our rally layer, not inside frozen detectors.\\n\\n**Option B (12?24+ months, after scaffolding + healthy corpus):** Collapse modular cues into a single owned multimodal rally model fine-tuned on (frames + audio + pose/stance + ?) ? rally [start, end]. Removes recall ceiling imposed by WASB candidate windows and licensing question on academic-trained weights. Requires: (a) enough golden eval/training video (owner self-labels via rally-annotator + vendor labels), (b) end-to-

end training harness (nanochat-style operability: one command, report card, deterministic, reproducible), (c) a measured improvement bar vs. the modular Option A.\n\n\n**Bridge data flow (cross-repo, ADR-002 contract):** sports-data-collector's per-match rally CSVs + manifest.json (commercial-license gate) ? indexer's golden eval set and training signal. Collector's 5-sport tables (badminton/tennis/cricket/pickleball/table-tennis) share a (video_id, ordinal, start_time, end_time) shape, so the bridge is sport-generic. Golden-set expansion funded via Rooru (Bengaluru) ? initial 3-sport pilot.\n\n\n**Immediate blockers on training-study readiness:** (1) **Owned-model base selection + licensing lock** (Qwen vs Gemma 4; verify per-checkpoint commercial terms). (2) **Fine-tuning backend decision** (self-managed ms-swift on rented GPU [RunPod/Lambda/Modal] vs. managed Together/Fireworks vs. enterprise Predibase). (3) **Golden QA eval harness design** (the crux: nanochat-style pass/fail verdict). (4) **Training data schema & consent** (how to convert rally CSVs + ending-reason labels + Gemini reference ? clip?QA pairs without using paid outputs; minors exclusion). (5) **Data leakage guardrails** (`train | eval` split by match, not clip; overlap detection). (6) **Training job orchestration** (how it slots into the async job model P0, runs on ComputeBackend, fits BYO-compute).\n\n\n**Success criteria for the study:** (a) Blueprint for converting existing golden labels into 5k?20k curated clip?QA pairs. (b) Confirmed Apache-2.0 base + verified commercial DPA from the fine-tune backend. (c) Proof-of-concept golden QA eval gate that blocks a regression-quality model. (d) Staged roadmap: Option A (modular, 2?4 weeks) ? Option B e2e model (after corpus health signals, 3?6 months post-scaffold). (e) Clear cost/latency comparison (self-hosted 7B VLM vs. Gemini per-call on target workload).\n\n",

"roadmap_alignment_notes": [

***Sequencing principle (owner directive):** Make scaffolding (P0?P4 in PLATFORM_ARCHITECTURE §6) rock-solid FIRST, keep the known paid stack (Gemini) as an always-ready fallback, and only then shift focus to owned-model quality (P5). This study unblocks P5's design but does NOT start implementation until P0?P4 infrastructure is delivered.",

***P0 (async job model, single-tenant)** is the prerequisite: owned-model fine-tuning runs as an orchestration job on ComputeBackend GPUs (§5A / DEFERRED #16). Without async + job dispatch, training is not operationalizable.",

***P1?P3 (multi-tenant, storage/compute abstraction)** enable the data flywheel: training corpus lives in Storage (§3.1), training/inference jobs dispatch to ComputeBackend (§3.2), per-user adapters live in the compute tier. Without these, the owned model is stranded on a single box.",

***P4 (rich rally/shot semantic metadata)** feeds P5: the annotation registry + candidate_segments + golden-set labels are the training-signal source. Per-rally ending_reason labels + human-correction loop ? QA-pair generation.",

***C14 (COMMERCIALIZATION.md)** tracks the owned-model licensing + training-data policy as a P1 blocker: Apache-2.0 base lock, open-model teachers only, no paid-LLM training signal, separate explicit training opt-in, minors excluded. This study must resolve C14 before any code lands.",

***ADR-004 substrate decision** (2026-06-02 update): Phase 4 offline segmenter is WASB shuttle-motion, not YOLO player-motion (YOLO @ IoU-0.5 recall too low). The owned end-to-end model (Option B) will eventually subsume WASB's role. ADR-007 audio fusion is the Option A stepping stone.",

***ADR-007 direction (accepted):** Audio hit-detection is the next rally-detection cue (Option A). It is fully local, permissive, and fuses cleanly with the distilled classifier. Gated on golden labels (E1 probe AMBER, awaits more data). This is the realistic 2?4 week Option A milestone while waiting for scaffold + corpus growth.",

***Golden-set expansion (ADR-006 / Rooru pilot):** Vendor-produced labels are the highest-leverage move. With Rooru on 3-sport badminton/pickleball/table-tennis, the training corpus grows month-on-month. The training study should plan for incremental re-training (CI-integrated, agent-runnable) as each batch lands.",

***Long-term moat (§5A, PLATFORM):** The defensible assets are (a) the consented, badminton-specific video?QA dataset and (b) the eval harness. Do NOT message 'we own an LLM' ? the base VLM (Qwen/Gemma) is freely downloadable. Message the dataset + eval precision instead.",

***Cost & margin (C7 / COMMERCIALIZATION):** Self-hosted 7B VLM (?16 GB VRAM) on rented GPU is cheaper per inference than Gemini only if: (a) sustained QA volume is high, (b) training corpus is defensibly large, (c) per-call API cost pain is real. Below those thresholds, paid API is cheaper than standing GPU + second stack. Benchmark before committing budget.",

***Training harness operability (§5A, PLATFORM):** The goal is 'an agent (or CI) can run a fine-tune on a fresh batch of golden-labeled video and get a trustworthy ship/no-ship verdict, unattended.' This requires: (1) one command, end-to-end (ms-swift engine, recipe-driven); (2) one complexity dial (auto-derived defaults); (3) automatic report card (held-out golden QA eval vs target); (4) reproducible + deterministic + cheap; (5) minimal/readable. Nanochat is the reference for operability (not the engine). The harness = thin glue over ms-swift + the golden

```
eval + the provider registry."
]
}
```

2. assistant (2026-06-08T13:04:41.932Z)

I'll research training compute platforms for the owned video-QA model. Let me start by loading the web tools and gathering current pricing and privacy data.

3. assistant (2026-06-08T13:04:44.967Z)

I'll research each platform systematically. Let me start with parallel searches on the key providers' pricing and privacy terms.

4. user (2026-06-08T13:04:52.395Z)

Web search results for query: "Modal GPU pricing 2026 A100 H100 per hour serverless"

Links: [{"title":"Plan Pricing | Modal","url":"https://modal.com/pricing"}, {"title":"Serverless GPU Pricing Matrix [2026 Developer Cheat Sheet]","url":"https://techbytes.app/posts/serverless-gpu-pricing-matrix-modal-replicate-lambda-2026/"}, {"title":"GPU Cloud Pricing 2026: H100 from \$1.03/hr, B200 from \$2.12/hr (15+ providers) | Spheron Blog","url":"https://www.spheron.network/blog/gpu-cloud-pricing-comparison-2026/"}, {"title":"Modal Pricing 2026: Serverless GPU & CPU Costs | CostBench","url":"https://costbench.com/software/ai-gpu-cloud/modal/"}, {"title":"Modal Pricing Calculator (2026) | BuildMVPFast","url":"https://www.buildmvpfast.com/tools/api-pricing-estimator/modal/"}, {"title":"Modal - Serverless GPU Provider & Pricing","url":"https://cloudgpuprices.com/vendors/modal/"}, {"title":"Modal Pricing and Alternatives: GPU vs. CPU Infrastructure | Blaxel Blog","url":"https://blaxel.ai/blog/modal-pricing-alternatives-guide/"}, {"title":"Every serverless GPU host compared: pricing, GPUs, and what they claim (April 2026) ? HostFleet","url":"https://hostfleet.net/serverless-gpu-pricing-matrix-2026/"}, {"title":"Modal Pricing Guide: Plans, Costs & Real Examples - Modal | CheckThat.ai","url":"https://checkthat.ai/brands/modal/pricing/"}, {"title":"GPU Cloud Pricing Comparison (June 2026) | RunPod, Lambda Labs, Modal","url":"https://www.buildmvpfast.com/api-costs/gpu"}]

Based on the search results, here's the current Modal GPU pricing for 2026:

Modal Serverless GPU Pricing (2026)

Per-Second Rates:

- H100: \$0.001097/sec
- A100 (80GB): \$0.000694/sec
- A100 (40GB): \$0.000583/sec

Hourly Equivalent:

- H100: approximately \$4.29/hr
- A100: approximately \$3.72/hr (80GB model)

Key Features:

- Modal is serverless, which means that we instantly autoscale up and down for you based on request volume.
- You only pay for compute time ? no idle charges.
- All new accounts get \$30/month in free compute credits.

Important Cost Modifiers:

- Regional multipliers range from 1.25x (US/EU/UK/Asia-Pacific) to 2.5x (other regions).
- Modal Sandboxes force non-preemptible pricing at \$0.00003942 per CPU core per second, a 3x premium over the base \$0.0000131 rate for general serverless functions.

Modal's per-second billing model makes it particularly cost-effective for bursty traffic, as you only pay for active compute time.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

5. user (2026-06-08T13:04:59.430Z)

Web search results for query: "RunPod GPU pricing H100 A100 on-demand spot 2026 data privacy terms"

Links: [{"title": "GPU Cloud Pricing | Per-Second H100, A100, RTX | Runpod", "url": "https://www.runpod.io/pricing"}, {"title": "GPU Cloud Pricing 2026: H100 from \$1.03/hr, B200 from \$2.12/hr (15+ providers) | Spheron Blog", "url": "https://www.spheron.network/blog/gpu-cloud-pricing-comparison-2026/"}, {"title": "RunPod GPU Pricing 2026 ? H100 & A100 Costs", "url": "https://gpubcost.org/provider/runpod"}, {"title": "RunPod Pricing 2026: Plans & Cost Analysis - RunPod | CheckThat.ai", "url": "https://checkthat.ai/brands/runpod/pricing"}, {"title": "Runpod GPU pricing: A complete breakdown and platform comparison | Blog ? Northflank", "url": "https://northflank.com/blog/runpod-gpu-pricing"}, {"title": "Cloud GPU Pricing 2026: A100 \$1.99/hr, H100 \$3.29/hr+ | SynpixonCloud", "url": "https://www.synpixoncloud.com/blog/cloud-gpu-pricing-comparison-2026"}, {"title": "GPU Cloud Pricing Comparison 2026: Lambda, RunPod, Vast.ai, AWS Live Rates", "url": "https://altstreet.investments/tools/gpu/gpu-price-comparison"}, {"title": "Runpod Secrets: Affordable A100/H100 Instances", "url": "https://www.runpod.io/articles/guides/affordable-a100-h100-gpu-cloud"}, {"title": "H100 Rental Prices Compared: \$1.49-\$6.98/hr Across 15+ Cloud Providers (2026) | IntuitionLabs", "url": "https://intuitionlabs.ai/articles/h100-rental-prices-cloud-comparison"}, {"title": "RunPod GPU Pricing: Compare 10+ GPUs | ComputePrices.com", "url": "https://computeprices.com/providers/runpod"}]

Now let me search for information about RunPod's data privacy terms:

Links: [{"title": "Data security and legal compliance - Runpod Documentation", "url": "https://docs.runpod.io/references/security-and-compliance"}, {"title": "Privacy Policy | Runpod", "url": "https://www.runpod.io/legal/privacy-policy"}, {"title": "AI Infrastructure Security & Compliance | Runpod", "url": "https://www.runpod.io/legal/compliance"}, {"title": "Keeping Data Secure: Best Practices for Handling Sensitive Data with Cloud GPUs", "url": "https://www.runpod.io/articles/guides/keep-data-secure-cloud-gpus"}, {"title": "docs/references/security-and-compliance.mdx at main · runpod/docs", "url": "https://github.com/runpod/docs/blob/main/references/security-and-compliance.mdx"}, {"title": "Runpod Achieves HIPAA and GDPR Compliance for AI Cloud Infrastructure", "url": "https://www.runpod.io/press/runpod-meets-hipaa-and-gdpr-standards"}, {"title": "Security Measures to Expect from AI Cloud Deployment Providers", "url": "https://www.runpod.io/articles/guides/security-measures-ai-cloud-deployment"}, {"title": "Data Processing Agreement", "url": "https://www.runpod.io/legal/data-processing-agreement"}, {"title": "Secure AI Deployments with Runpod's SOC2 Compliance", "url": "https://www.runpod.io/articles/guides/secure-ai-deployments-soc2-compliance"}]

Summary of RunPod GPU Pricing and Data Privacy (2026)

****GPU Pricing Overview****

****H100 Pricing (2026):****

- RunPod's H100 costs \$2.65/hr
- RunPod spot occasionally hits \$1.19/hr depending on regional availability
- Spheron undercuts RunPod on H100 pricing (\$2.01 vs \$3.29 on-demand)

****A100 Pricing (2026):****

- RunPod's A100 pricing starts at \$1.29/hr, ideal for projects with tighter budgets or smaller-scale workloads
- Spheron offers A100 80GB at \$1.07/hr on-demand and \$0.60/hr spot

****Pricing Context:****

- RunPod's A100 pricing is consistent at \$1.19/hr, whereas AWS spot instances range from \$0.41-\$1.23/hr with ~5% interruption rates
- Spot/preemptible instances can offer 50-90% savings but come with the risk of interruption

****Data Privacy and Security Terms****

****Compliance Certifications:****

- RunPod has been independently audited and verified as meeting HIPAA and GDPR standards, enabling healthcare organizations and EU companies to process sensitive data with enhanced security and privacy protections
- RunPod is SOC 2 Type II compliant, which means it has been audited for security controls to protect customer data
- Secure Cloud provides vetted infrastructure partners who meet enterprise security standards including SOC 2, ISO 27001, and PCI DSS certifications

****Data Isolation and Access:****

- Pods and workers run in a multi-tenant environment with containerized isolation; each Pod/worker operates in its own container with strict separation from other workloads
- RunPod's terms of service prohibit hosts from inspecting Pod/worker data or analyzing usage patterns, with violations resulting in immediate removal from the platform

****Encryption and Security Measures:****

- Robust encryption protocols ensure that all data, whether in transit between the platform and data centers or during processing, is kept confidential and secure
- All data on RunPod is AES-256 encrypted at rest and secured using TLS in transit

****GDPR Compliance:****

- The platform facilitates data subject rights under GDPR, including the right to access, rectify, erase, or restrict the processing of personal data, with requests handled promptly and efficiently

****Secure Cloud Option:****

- For sensitive workloads requiring enhanced security, Secure Cloud operates in T3/T4 data centers with enterprise-grade security, high reliability, redundancy, and fast response times

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-08T13:05:02.086Z)

Let me continue with the major cloud providers and the remaining serverless/independent GPU clouds.

7. user (2026-06-08T13:05:09.402Z)

Web search results for query: "Lambda Labs GPU cloud pricing H100 A100 on-demand 2026 data privacy retention"

Links: [{"title":"Lambda Cloud H100 Pricing 2026: Per-Hour Cost Breakdown and Spheron Comparison | Spheron Blog","url":"https://www.spheron.network/blog/lambda-cloud-h100-pricing-2026/"},"{"title":"AI Cloud Pricing | GPU Compute & AI Infrastructure | Lambda","url":"https://lambda.ai/pricing"},"{"title":"Lambda Labs Pricing 2026: Plans, Costs & Comparison - Lambda Labs | CheckThat.ai","url":"https://checkthat.ai/brands/lambda-labs/pricing"},"{"title":"GPU Cloud Pricing Comparison 2026: Lambda, RunPod, Vast.ai, AWS Live Rates","url":"https://altstreet.investments/tools/gpu/gpu-price-comparison"},"{"title":"Lambda Labs H100 Price: \$2.99/hr (Apr 2026) | SynpixonCloud","url":"https://www.synpixoncloud.com/blog/lambda-labs-gpu-pricing-2026"},"{"title":"Lambda Labs GPU Pricing: 11 GPUs from \$0.69/hr (2026)","url":"https://gpuperhour.com/providers/lambda-labs"},"{"title":"Lambda Labs GPU Pricing: Compare 10+ GPUs | ComputePrices.com","url":"https://computeprices.com/providers/lambda"},"{"title":"Lambda Labs

Alternatives: Find low-price A100 and H100 (2026) | Thunder Compute", "url": "https://www.thundercompute.com/blog/lambda-labs-alternatives"}, {"title": "RunPod vs Lambda Labs vs CoreWeave: GPU Cloud Pricing", "url": "https://www.buildmvpfast.com/blog/gpu-cloud-cost-comparison-runpod-lambda-labs-coreweave-2026"}, {"title": "Best Lambda Labs Alternatives for H100 GPU Rental in 2026", "url": "https://www.gmicloud.ai/en/blog/best-lambda-labs-alternatives-for-renting-h100-gpus-in-2026"}]]

Based on the search results, here's what I found about Lambda Labs GPU cloud pricing for H100 and A100 on-demand instances in 2026:

Lambda Labs H100 On-Demand Pricing

Lambda Cloud H100 on-demand runs \$3.29-\$3.99/hr. More specifically:
- H100 SXM now starts at \$4.29/hr for a single GPU and falls to \$3.99/GPU/hr in 8-GPU configs, while H100 PCIe is \$3.29/hr

Lambda Labs A100 On-Demand Pricing

Lambda Labs charges \$2.49/hr for A100 80GB on-demand instances as of March 2026. Additionally, A100 80GB SXM in 8-GPU systems is \$2.79/GPU/hr, while A100 PCIe instances are listed at \$1.99/hr.

Key Features

There is no spot or preemptible pricing, and data transfer out is free, unlike AWS (\$0.09/GB) and GCP (\$0.12/GB).

****Note:**** I did not find specific information about data privacy retention policies in the search results. For detailed privacy and data retention information, I'd recommend checking Lambda Labs' official documentation or contacting their support directly.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. user (2026-06-08T13:05:12.974Z)

Web search results for query: "Google Vertex AI custom training GPU pricing A100 H100 India asia-south1 region data privacy not used for training"

Links: [{"title": "Vertex AI pricing", "url": "https://cloud.google.com/vertex-ai/pricing"}, {"title": "Cloud GPU Pricing Comparison India [2026]: A100, H200 & More", "url": "https://acecloud.ai/blog/cloud-gpu-pricing-comparison/"}, {"title": "VM instance pricing | Google Cloud", "url": "https://cloud.google.com/compute/gpus-pricing"}, {"title": "Vertex AI Pricing: The Complete 2026 Guide to Costs, Hidden Fees, and Savings", "url": "https://www.nops.io/blog/vertex-ai-pricing/"}, {"title": "Google Cloud A3 H100 Pricing 2026: GCP Per-Hour Cost vs Spheron Breakdown | Spheron Blog", "url": "https://www.spheron.network/blog/google-cloud-a3-h100-pricing/"}, {"title": "Google Vertex AI Pricing: Complete Enterprise Guide (2026)", "url": "https://www.cloudzero.com/blog/google-vertex-ai-pricing/"}, {"title": "How to Use Vertex AI Training with Reserved GPU Clusters", "url": "https://oneuptime.com/blog/post/2026-02-17-how-to-use-vertex-ai-training-with-reserved-gpu-clusters-for-predictable-workloads/view"}, {"title": "GPU Cloud Pricing 2026: H100 From \$2.21/hr ? AWS vs Azure vs GCP Compared", "url": "https://spendark.com/blog/machine-learning-cloud-cost/"}, {"title": "Google Vertex AI Pricing: Complete 2026 Price Guide | DeployBase", "url": "https://deploybase.ai/articles/vertex-ai-pricing"}, {"title": "SKU Groups - Vertex Training and Prediction (H100) | Google Cloud", "url": "https://cloud.google.com/skus/sku-groups/vertex-training-and-prediction-h100"}]]

I found search results with pricing information for Google Cloud GPUs, but let me search specifically for the India region (asia-south1) and data privacy policies.

Links: [{"title": "How should I manage data that is not used for training in Google Vertex AI? -

UMU", "url": "https://www.umu.com/ask/q11122301573854289934"}, {"title": "What happens to data in Google Vertex AI that is not used for training? -
UMU", "url": "https://www.umu.com/ask/q11122301573854289928"}, {"title": "Does Claude Train on Your Data?", "url": "https://fonzi.ai/blog/claude-privacy-and-data-training"}, {"title": "Updates to Consumer Terms and Privacy Policy \\ Anthropic", "url": "https://www.anthropic.com/news/updates-to-our-consumer-terms"}, {"title": "Google Cloud Whitepaper September 2024 Generative AI, Privacy, and Google Cloud", "url": "https://services.google.com/fh/files/misc/genai_privacy_google_cloud_202308.pdf"}, {"title": "Google Cloud generative AI data residency guarantees for data stored at-rest | Google Cloud Blog", "url": "https://cloud.google.com/blog/products/ai-machine-learning/google-cloud-generative-ai-data-residency-guarantees-for-data-stored-at-rest"}, {"title": "SoK: The Gap Between Data Rights Ideals and Reality", "url": "https://arxiv.org/pdf/2312.01511"}, {"title": "How Sensitive Data Protection can help secure generative AI workloads | Google Cloud Blog", "url": "https://cloud.google.com/blog/products/identity-security/how-sensitive-data-protection-can-help-secure-generative-ai-workloads"}, {"title": "\"We are not Future-ready\": Understanding AI Privacy Risks and Existing Mitigation Strategies from the Perspective of AI Developers in Europe", "url": "https://arxiv.org/pdf/2510.00909"}]

Links: [{"title": "Cloud GPU Pricing Comparison India [2026]: A100, H200 & More", "url": "https://acecloud.ai/blog/cloud-gpu-pricing-comparison/"}, {"title": "VM instance pricing | Google Cloud", "url": "https://cloud.google.com/compute/gpus-pricing"}, {"title": "Google Cloud A3 H100 Pricing 2026: GCP Per-Hour Cost vs Spheron Breakdown | Spheron Blog", "url": "https://www.spheron.network/blog/google-cloud-a3-h100-pricing/"}, {"title": "a3-highgpu-8g specs and pricing | GCP | CloudPrice", "url": "https://cloudprice.net/gcp/compute/instances/a3-highgpu-8g"}, {"title": "Accelerator-optimized VM Pricing | Google Cloud", "url": "https://cloud.google.com/products/compute/pricing/accelerator-optimized"}, {"title": "NVIDIA H100 Price in India: Complete Cloud vs Purchase Guide (2025) | E2E Networks", "url": "https://www.e2enetworks.com/blog/nvidia-h100-price-india"}, {"title": "Pricing | Spot VMs | Google Cloud", "url": "https://cloud.google.com/spot-vms/pricing"}, {"title": "GPU Cloud Pricing: India vs Global Providers ? Full Cost Comparison (2026)", "url": "https://cyfuture.ai/blog/gpu-cloud-pricing-in-india-vs-global-providers"}, {"title": "GPU locations | Compute Engine | Google Cloud Documentation", "url": "https://docs.cloud.google.com/compute/docs/regions-zones/gpu-regions-zones"}, {"title": "H100 GPU Pricing in India | Gpu | Cloud Computing Dictionary | E2E Networks", "url": "https://www.e2enetworks.com/cloud-terms/gpu/h100-pricing-india"}]

Based on my web search results, here's what I found regarding your query about Google Vertex AI custom training GPU pricing, regional availability, and data privacy:

GPU Pricing and Regional Availability

****India Region (asia-south1) Availability:****

The a3-highgpu-8g (8 NVIDIA H100 GPU configuration) is available in 16 regions starting from \$64,117.70 per month, including India's asia-south1 region.

****A100 and H100 Pricing Context:****

A single A100 GPU costs \$2.93 per hour in us-central1, and charges accumulate from the moment Google provisions resources until the job completes. Additionally, Vertex AI adds management fees on top of underlying Compute Engine costs?an NVIDIA A100 GPU costs \$2.93 per hour for compute plus \$0.44 per hour Vertex management fee, totaling \$3.37 per hour.

An H100 at \$11.27/hour means a 24-hour distributed training run on eight GPUs costs roughly \$2,164.

****Regional Pricing Variations:****

Regional pricing for europe-west4 and asia-southeast1 typically runs 7-15% above us-central1 rates. This suggests asia-south1 pricing may also carry a regional premium.

Data Privacy Policy

****Key Data Privacy Commitment:****

Customer data is never used by Google. Google does not use customer data to train their models, or models used by others, without permission.

This commitment applies to Vertex AI custom training jobs?your training data is not used to improve Google's models unless you explicitly authorize it.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-06-08T13:05:16.218Z)

Let me research the remaining providers: AWS SageMaker, Azure ML, vast.ai, CoreWeave, Together AI, Paperspace, plus the self-hosted RTX 2070 Super option for fine-tuning a 7B VLM.

10. user (2026-06-08T13:05:28.625Z)

Web search results for query: "vast.ai GPU pricing H100 A100 2026 data privacy security marketplace decentralized risks"

Links: [{"title": "GPU Pricing ? Live Platform Rates | Vast.ai", "url": "https://vast.ai/pricing"}, {"title": "GPU Cloud Pricing 2026: H100 from \$1.03/hr, B200 from \$2.12/hr (15+ providers) | Spheron Blog", "url": "https://www.spheron.network/blog/gpu-cloud-pricing-comparison-2026/"}, {"title": "GPU Cloud Pricing Comparison 2026: Lambda, RunPod, Vast.ai, AWS Live Rates", "url": "https://altstreet.investments/tools/gpu/gpu-price-comparison"}, {"title": "Rent H100 PCIE GPUs on Vast.ai for \$2.13/hr", "url": "https://vast.ai/pricing/gpu/H100-PCIE"}, {"title": "Vast.ai GPU Pricing: Compare 10+ GPUs | ComputePrices.com", "url": "https://computeprices.com/providers/vast"}, {"title": "Rent H100 NVL GPUs on Vast.ai for \$2.00/hr", "url": "https://vast.ai/pricing/gpu/H100-NVL"}, {"title": "GPU Cloud Hosting for AI & ML | Vast.ai", "url": "https://vast.ai/products/gpu-cloud"}, {"title": "H100 GPU Cost In 2026: Buy, Rent, And Cloud Pricing Compared", "url": "https://www.cloudzero.com/blog/h100-gpu-cost/"}, {"title": "H100 Rental Prices Compared: \$1.49-\$6.98/hr Across 15+ Cloud Providers (2026) | IntuitionLabs", "url": "https://intuitionlabs.ai/articles/h100-rental-prices-cloud-comparison"}, {"title": "Cloud GPU Pricing 2026: A100 \$1.99/hr, H100 \$3.29/hr+ | SynpixonCloud", "url": "https://www.synpixoncloud.com/blog/cloud-gpu-pricing-comparison-2026"}]

Links: [{"title": "Compliance | Vast.ai", "url": "https://vast.ai/compliance"}, {"title": "Running Private AI Models Without the Risk of Data Exposure", "url": "https://vast.ai/article/running-private-ai-models-without-the-risk-of-data-exposure"}, {"title": "Security FAQ - Vast.ai Documentation: Affordable GPU Cloud Marketplace", "url": "https://docs.vast.ai/guides/reference/faq/security"}, {"title": "In-Depth Analysis of Vast.ai: The Cloud Computing Platform AI Users Can't Ignore", "url": "https://skywork.ai/skypage/en/In-Depth-Analysis-of-Vast.ai-The-Cloud-Computing-Platform-AI-Users-Can-t-Ignore/1972586731313754112"}, {"title": "Privacy Policy | Vast.ai", "url": "https://vast.ai/privacy"}, {"title": "Trust Center - Vast.ai", "url": "https://trust.vast.ai/"}, {"title": "VAST Data and CrowdStrike Partner to Secure the AI Lifecycle", "url": "https://www.crowdstrike.com/en-us/press-releases/vast-data-crowdstrike-partner-secure-ai-lifecycle/"}, {"title": "Vast.ai Alternatives: 66 Open-Source & API Options (2026)", "url": "https://infrabase.ai/alternatives/vast-ai"}, {"title": "Is Vast AI Safe? Learn if Vast AI Is Legit | Nudge Security", "url": "https://www.nudgesecurity.com/security-profile/vast-ai"}, {"title": "VAST AI Operating System: Powering the Agentic AI Revolution - VAST Data", "url": "https://www.vastdata.com/"}]

Links: [{"title": "Running Private AI Models Without the Risk of Data Exposure", "url": "https://vast.ai/article/running-private-ai-models-without-the-risk-of-data-exposure"}, {"title": "In-Depth Analysis of Vast.ai: The Cloud Computing Platform AI Users Can't Ignore", "url": "https://skywork.ai/skypage/en/In-Depth-Analysis-of-Vast.ai-The-Cloud-Computing-Platform-AI-Users-Can-t-Ignore/1972586731313754112"}, {"title": "Vast.ai Review: GPU Rental Prices and Real Tradeoffs 2026", "url": "https://www.aitooldiscovery.com/ai-infra/vast-ai-review"}, {"title": "Security FAQ - Vast.ai Documentation: Affordable GPU Cloud Marketplace", "url": "https://docs.vast.ai/guides/reference/faq/security"}, {"title": "Rent GPUs | Vast.ai", "url": "https://vast.ai/"}, {"title": "Vast.ai GPU Hosting Wiki ? Crypto Labs (Pty) Ltd", "url": "https://www.cryptolabs.co.za/vast-ai-gpu-hosting-wiki/"}, {"title": "Why just not rent a 3090 from vast ai? This is literally 0.2\$ per hour. | Hacker News", "url": "https://news.ycombinator.com/item?id=36026101"}]

I've completed a comprehensive web search on Vast.ai covering GPU pricing, data privacy,

security, marketplace characteristics, and associated risks. Here's a summary of the key findings:

****GPU Pricing (H100 & A100) - 2026****

Vast.ai offers H100s for as little as \$0.90/hour and saves up to 80% compared to AWS, Azure, or GCP. More specifically, Vast.ai marketplace pricing for A100 80GB dips to \$0.67/hr when high-reliability hosts are available, but consistency varies host-to-host. Vast.ai marketplace H100 pricing is around \$1.87, though spot pricing on Vast.ai goes as low as \$0.34/hr with interruption risk.

****Data Privacy & Security****

Vast.ai complies with applicable US state privacy laws, including CCPA/CPRA, and outlines data collection, use, retention, and deletion practices in their Privacy Policy. Every workload runs in an unprivileged Docker container isolated from other tenants, clients access only their own data with no shared filesystems between tenants, and data is destroyed immediately when a client deletes an instance.

Vast.ai has completed SOC 2 Type 2 certification, which verified that security, availability, and confidentiality controls meet AICPA Trust Services Criteria.

****Marketplace & Decentralized Model****

Prices are set by supply and demand across 40+ data centers, with on-demand, interruptible, or reserved options available. Vast.ai operates on a peer-to-peer model, which comes with a fundamental trade-off between the low cost of hobbyist machines and the higher reliability of vetted data centers.

****Risks & Security Concerns****

The peer-to-peer nature introduces inherent security risks compared to a centralized provider. SOC2 certification does not cover individual hosts' physical security or network configuration, and hosts are not themselves audited to SOC2 standards. Price volatility is a concern, as a host can change pricing at any time?instances launched at \$0.29/hr may cost \$0.39/hr after the host updates their listing.

For stricter security, Vast.ai's Secure Cloud tier restricts infrastructure to vetted datacenter partners holding a minimum of ISO 27001 certification, with many also HIPAA, NIST, PCI, and/or SOC 1-3 certified, as well as GDPR compliant.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

11. user (2026-06-08T13:05:31.147Z)

Web search results for query: "AWS SageMaker training ml.p4d ml.p5 H100 A100 pricing 2026 India ap-south1 Mumbai region data privacy"

Links: [{"title": "SageMaker pricing - AWS", "url": "https://aws.amazon.com/sagemaker/pricing/"}, {"title": "Amazon SageMaker AI Pricing", "url": "https://aws.amazon.com/sagemaker/ai/pricing/"}, {"title": "AWS GPU Instance Pricing: P5, P4d, G5, Inf2 | Wring Blog", "url": "https://wring.co/blog/aws-gpu-instance-pricing-guide"}, {"title": "Amazon SageMaker Pricing Guide: 2026 Costs (And Savings)", "url": "https://www.cloudzero.com/blog/sagemaker-pricing/"}, {"title": "ml.p5.4xlarge specs and pricing | Amazon SageMaker AI | CloudPrice", "url": "https://cloudprice.net/aws/sagemake r/instances/ml.p5.4xlarge"}, {"title": "SageMaker Pricing - ML Instance Types & GPU Pricing | CloudPrice", "url": "https://cloudprice.net/aws/sagemaker"}, {"title": "AWS SageMaker Pricing: The Developer?s Guide to Smart ML Spending", "url": "https://www.cloudexmachina.io/blog/sagemaker-pricing"}, {"title": "SageMaker Pricing 2026: Plans, Costs & Hidden Fees - Amazon SageMaker | CheckThat.ai", "url": "https://checkthat.ai/brands/amazon-sagemaker/pricing"}, {"title": "Amazon SageMaker AI Pricing: Detailed Breakdown and Ultimate

Guide", "url": "https://cloudchipr.com/blog/amazon-sagemaker-pricing"}, {"title": "Holori - Ultimate AWS SageMaker pricing guide", "url": "https://holori.com/aws-sagemaker-pricing/"}]

I found some general SageMaker pricing information, but the results don't specifically cover ap-south-1 (Mumbai) region pricing or data privacy details. Let me search more specifically for India region availability and data privacy.

Links: [{"title": "Supported Regions and Quotas - Amazon SageMaker AI", "url": "https://docs.aws.amazon.com/sagemaker/latest/dg/regions-quotas.html"}, {"title": "Actual availability of instance type in region | AWS re:Post", "url": "https://repost.aws/questions/QUcT6yodj9RtqfvoQn04ABfg/actual-availability-of-instance-type-in-region"}, {"title": "Instance Types ? sagemaker 2.224.1 documentation", "url": "https://sagemaker.readthedocs.io/en/v2.224.1/api/utility/instance_types.html"}, {"title": "Amazon SageMaker AI - AWS Services and Regions", "url": "https://www.aws-services.info/sagemaker.html"}, {"title": "Docker Registry Paths and Example Code for Asia Pacific (Mumbai) (ap-south-1) - Amazon SageMaker", "url": "https://docs.aws.amazon.com/sagemaker/latest/dg/ecr-ap-south-1.html"}, {"title": "Amazon SageMaker AI Region Availability - Available in 37 AWS Regions! --> | AWS Fundamentals", "url": "https://awsfundamentals.com/regions/service/amazon-sagemaker-ai"}, {"title": "How do I resolve the Amazon SageMaker AI endpoint error ? Unable to locate at least 2 availability zone(s) with the requested instance type that overlap with SageMaker subnets?? | AWS re:Post", "url": "https://repost.aws/articles/AR-hzmpfX1QBONBC5SheQdg/how-do-i-resolve-the-amazon-sage-maker-ai-endpoint-error-unable-to-locate-at-least-2-availability-zone-s-with-the-requested-instance-type-that-overlap-with-sage-maker-subnets"}, {"title": "Not able to create t4g.small ec2 type instance in mumbai local zone | AWS re:Post", "url": "https://repost.aws/questions/QU1UJva0kgSiGOCPCZqVdWeA/not-able-to-create-t4g-small-ec2-type-instance-in-mumbai-local-zone"}, {"title": "Amazon SageMaker AI endpoints and quotas - AWS General Reference", "url": "https://docs.aws.amazon.com/general/latest/gr/sagemaker.html"}]

Links: [{"title": "Data Protection in Amazon SageMaker AI - Amazon SageMaker AI", "url": "https://docs.aws.amazon.com/sagemaker/latest/dg/data-protection.html"}, {"title": "Data protection - SageMaker Studio Administration Best Practices", "url": "https://docs.aws.amazon.com/whitepapers/latest/sagemaker-studio-admin-best-practices/data-protection.html"}, {"title": "Configure security in Amazon SageMaker AI - Amazon SageMaker AI", "url": "https://docs.aws.amazon.com/sagemaker/latest/dg/security.html"}, {"title": "Data Privacy in Amazon SageMaker AI - Amazon SageMaker AI", "url": "https://docs.aws.amazon.com/sagemaker/latest/dg/data-privacy.html"}, {"title": "General Data Protection Regulation (GDPR) Center", "url": "https://aws.amazon.com/compliance/gdpr-center/"}, {"title": "Top GDPR Compliance Software for Amazon SageMaker in 2026", "url": "https://slashdot.org/software/gdpr-compliance/for-amazon-sagemaker/"}, {"title": "AWS Service Terms - Amazon.com", "url": "https://aws.amazon.com/service-terms/"}, {"title": "Securing Machine Learning Pipelines: Best Practices in Amazon SageMaker", "url": "https://tutorialsdodojo.com/securing-machine-learning-pipelines-best-practices-in-amazon-sagemaker/"}, {"title": "Data protection in Amazon SageMaker Unified Studio - Amazon SageMaker Unified Studio", "url": "https://docs.aws.amazon.com/sagemaker-unified-studio/latest/adminguide/data-protection.html"}]

Based on my web search results, here's what I found regarding your comprehensive query about AWS SageMaker training with ml.p4d/ml.p5 GPU instances in the India ap-south-1 (Mumbai) region:

Regional Availability

Amazon SageMaker AI is available in 37 of 37 AWS regions (100%), which includes ap-south-1 (Mumbai). The region supports SageMaker services and has Docker registry paths configured for various frameworks.

Pricing Information

AWS announced up to 45% price reductions on P4 and P5 GPU instances effective June 2025. The search results indicate that ml.p4d.24xlarge instances (with 8x A100 GPUs) are available at competitive rates for training large models, though specific ap-south-1 pricing wasn't detailed in these results.

SageMaker instance costs are billed by the hour (or second, with a 1-minute minimum), and you can achieve significant savings through Spot instances and Savings Plans.

Data Privacy & Compliance

SageMaker offers comprehensive data protection features:

- You can pass your AWS KMS key to encrypt an EBS volume attached to training jobs, and if you don't specify a KMS key, SageMaker AI encrypts both operating system volumes and ML data volumes with a system-managed KMS key. Sensitive data that needs to be encrypted with a KMS key for compliance reasons should be stored in the ML storage volume or in Amazon S3.
- By default, Amazon SageMaker AI runs training jobs in an Amazon VPC, and you can configure a private VPC and set up VPC endpoints to route traffic over a private network without egressing traffic over the internet.
- AWS customers can use all AWS services to process personal data (as defined in the GDPR) that is uploaded to the AWS services under their AWS accounts in compliance with the GDPR.
- Amazon SageMaker AI collects aggregate information about the use of AWS-owned and open source libraries used during training, and uses this aggregate metadata to improve services and customer experience, with options to opt out of sharing aggregated metadata when creating a training job.

For the most current and region-specific pricing for ml.p4d and ml.p5 instances in ap-south-1, I recommend checking the AWS SageMaker pricing page directly or using AWS pricing tools.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

12. assistant (2026-06-08T13:05:34.120Z)

Let me research the remaining providers: CoreWeave, Together AI, Paperspace, Azure ML, and the critical self-hosted RTX 2070 Super feasibility for 7B VLM fine-tuning.

13. user (2026-06-08T13:05:44.531Z)

Web search results for query: "Together AI fine-tuning pricing 2026 data privacy does not train on customer data DPA LoRA"

Links: [{"title": "Fine-Tuning | Together AI", "url": "https://www.together.ai/fine-tuning"}, {"title": "A complete guide to Together AI pricing in 2026 | eesel AI", "url": "https://www.eesel.ai/blog/together-ai-pricing"}, {"title": "Together.ai API Pricing 2026 ? All Models, Live Rates | AI Pricing Guru", "url": "https://www.aipricing.guru/together-pricing/"}, {"title": "Pricing - Together AI docs", "url": "https://docs.together.ai/docs/fine-tuning-pricing"}, {"title": "LoRA Fine-Tuning Cost 2026 | Real Pricing + ROI Calculator - Stratagem Systems", "url": "https://www.stratagem-systems.com/blog/lora-fine-tuning-cost-analysis-2026"}, {"title": "Pricing | Together AI", "url": "https://www.together.ai/pricing"}, {"title": "LLM Fine-Tuning Pricing 2026 ? Compare Training Costs | Price Per Token", "url": "https://pricepertoken.com/fine-tuning"}, {"title": "Together AI Pricing 2026: Plans, Costs & Rates - Together AI | CheckThat.ai", "url": "https://checkthat.ai/brands/together-ai/pricing"}, {"title": "Fine-Tuning Costs Comparison - Train Your Own AI | Awesome Agents", "url": "https://awesomeagents.ai/pricing/fine-tuning-costs-comparison/"}, {"title": "Products | Together AI", "url": "https://www.together.ai/products"}]

Based on the search results, here's what I found about Together AI's fine-tuning pricing, LoRA support, and data privacy:

Pricing Information for 2026

LoRA SFT on a 16B base is \$0.48/M tokens; full fine-tuning on a 70-100B base is \$3.20/M; DPO on a 70-100B is \$8.00/M. More specifically, LoRA fine-tuning is priced at \$0.48/M tokens for models

up to 16B parameters, and typical training cost is around \$8-\$12 per million training tokens depending on model size, with hosted serving of the fine-tuned model at standard rates plus a small LoRA overhead.

LoRA Support

The new ability to resume from a checkpoint combined with LoRA serving has enabled our customers to deeply tune our foundation model, ShieldLlama, for their enterprise's precise risk posture. LoRA fine-tuning is the killer feature. Together AI supports fine-tuning on every major Llama, Mistral, and Qwen size including the 405B flagship, at roughly \$8-\$12 per million training tokens.

Data Privacy

Regarding data privacy, Together AI takes security and compliance seriously, with strict data privacy controls to keep your information protected. Your data and models remain fully under your ownership, safeguarded by robust security measures.

However, the search results did not contain specific information about data processing agreements (DPA) or explicit statements confirming that Together AI does not train on customer data. You may need to review their official documentation or contact their sales team for detailed information about DPA terms and data processing policies.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

14. user (2026-06-08T13:05:50.853Z)

Web search results for query: "CoreWeave Paperspace H100 A100 pricing 2026 minimum commitment data privacy"

Links: [{"title":"CoreWeave Pricing Guide (June 2026)","url":"https://www.thundercompute.com/blog/coreweave-gpu-pricing-review"}, {"title":"10 Best CoreWeave Alternatives in 2026: Cheaper GPUs Without the Lock-In | Spheron Blog","url":"https://www.spheron.network/blog/coreweave-alternatives/"}, {"title":"CoreWeave Cloud Pricing | CoreWeave","url":"https://www.coreweave.com/pricing"}, {"title":"CoreWeave Classic Pricing | CoreWeave","url":"https://www.coreweave.com/pricing/classic"}, {"title":"H100 Rental Prices Compared: \$1.49-\$6.98/hr Across 15+ Cloud Providers (2026) | IntuitionLabs","url":"https://intuitionlabs.ai/articles/h100-rental-prices-cloud-comparison"}, {"title":"CoreWeave Pricing 2026: Plans, Costs & What You'll Pay - CoreWeave | CheckThat.ai","url":"https://checkthat.ai/brands/coreweave/pricing"}, {"title":"CoreWeave | Review, Pricing & Alternatives","url":"https://getdeploying.com/coreweave"}, {"title":"CoreWeave pricing 2026: GPU costs per hour explained","url":"https://www.eesel.ai/blog/coreweave-pricing"}, {"title":"CoreWeave GPU Pricing: 9 GPUs from \$1.19/hr (2026)","url":"https://gpuperhour.com/providers/coreweave"}, {"title":"CoreWeave Pricing 2026: 4 Plans from \$10-\$68.8/instance/hour","url":"https://costbench.com/software/ai-gpu-cloud/coreweave/"}]

Based on the search results, here's what I found regarding CoreWeave, Paperspace, H100, A100 pricing, minimum commitments, and data privacy for 2026:

Pricing (2026)

****CoreWeave:****

- H100 pricing works out to about \$6.16/hr per GPU when normalized from \$49.24/hr 8-GPU HGX nodes
- A100 pricing works out to about \$2.70/hr per GPU when normalized from 8-GPU nodes

****Paperspace:****

- H100 pricing sits at \$5.95 per hour (promotional rate)
- A100 GPUs run \$3.09 per hour

Minimum Commitment

- Users pay only for what they use, can scale clusters up or down in real time, and avoid long-term hardware commitments for CoreWeave's on-demand model
- Long-term reserved contracts standard for enterprise customers; on-demand and spot available without commitment
- Paperspace has no minimum commitment, but you pay for rented time regardless

Data Privacy

CoreWeave holds SOC 2, HIPAA, GDPR, ISO 27001 certifications, indicating strong data privacy and security compliance standards.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

15. assistant (2026-06-08T13:05:53.735Z)

Now let me research Azure ML and the critical self-hosted RTX 2070 Super feasibility question for fine-tuning a 7B VLM.

16. user (2026-06-08T13:06:01.444Z)

Web search results for query: "Azure Machine Learning GPU pricing NC A100 H100 2026 Central India region data privacy not used for training"

Links: [{"title": "Pricing - Azure Machine Learning | Microsoft Azure", "url": "https://azure.microsoft.com/en-us/pricing/details/machine-learning/"}, {"title": "GPU Cloud Pricing 2026: H100 From \$2.21/hr ? AWS vs Azure vs GCP Compared", "url": "https://spendark.com/blog/machine-learning-cloud-cost/"}, {"title": "Cloud GPU Pricing Comparison India [2026]: A100, H200 & More", "url": "https://acecloud.ai/blog/cloud-gpu-pricing-comparison/"}, {"title": "GPU Cloud Pricing 2026: H100 from \$1.03/hr, B200 from \$2.12/hr (15+ providers) | Spheron Blog", "url": "https://www.spheron.network/blog/gpu-cloud-pricing-comparison-2026/"}, {"title": "Cloud GPU Pricing Comparison: AWS Vs Azure Vs GCP For AI Workloads (2026)", "url": "https://www.cloudzero.com/blog/cloud-gpu-pricing-comparison/"}, {"title": "NC24ads A100 v4 pricing and specs - Vantage Instances", "url": "https://instances.vantage.sh/azure/vm/nc24ads-v4"}, {"title": "NC_A100_v4 size series - Azure Virtual Machines | Microsoft Learn", "url": "https://learn.microsoft.com/en-us/azure/virtual-machines/sizes/gpu-accelerated/nc100v4-series"}, {"title": "GPU Price 2025 Report ? Cloud GPU Cost & Trends", "url": "https://cast.ai/reports/gpu-price/"}, {"title": "Standard_NC24ads_A100_v4 specs and pricing | Azure | CloudPrice", "url": "https://cloudprice.net/vm/Standard_NC24ads_A100_v4"}, {"title": "Pricing Calculator | Microsoft Azure", "url": "https://azure.microsoft.com/en-us/pricing/calculator/"}]

Based on the search results, here's information about Azure Machine Learning GPU pricing for NC A100 and H100 models in 2026:

Azure GPU Pricing (2026)

Azure NC H100 v5 is priced at \$98.46/hr for on-demand usage. For the A100 series, the NC24ads A100 v4 starts at \$3.673 per hour on-demand or \$0.821283 per hour with spot instances.

Azure's single-GPU NC H100 v5 runs around \$6.98 per hour, while its 8-GPU ND H100 v5 instance is closer to \$12 per GPU per hour, with pricing varying by region and some locations exceeding \$9 to \$10 per GPU per hour.

Central India Region Pricing

Regarding the Central India region specifically, Azure pricing for an 8× H100 setup is approximately ₹38.6 lakh (Indian Rupees) as of early 2026.

Data Privacy Note

Regarding your question about data privacy and data not being used for training: The search results do not contain specific information about Azure Machine Learning's data privacy policies or guarantees about data not being used for model training in the Central India region. For detailed information on these policies, you would need to consult Microsoft's official Azure Machine Learning documentation and privacy agreements.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

17. user (2026-06-08T13:06:02.428Z)

Web search results for query: "fine-tune Qwen2.5-VL 7B QLoRA single 8GB GPU RTX 2070 VRAM requirements possible"

Links: [{"title": "Qwen 2.5-VL 7B Fine-Tuning Guide for Segmentation", "url": "https://www.labellerr.com/blog/fine-tune-qwen-2-5-vl/"}, {"title": "Fine-Tuning a Vision Language Model (Qwen2-VL-7B) with the Hugging Face Ecosystem (TRL) · Hugging Face", "url": "https://huggingface.co/learn/cookbook/fine_tuning_vlm_trl"}, {"title": "Fine-Tune and Deploy Qwen2.5 - Coder with SFT & DPO on PAI - Alibaba Cloud", "url": "https://www.alibabacloud.com/help/en/pai/use-cases/training-evaluation-compression-and-deployment-of-qwen2-5-coder-model"}, {"title": "Qwen2.5 QLoRA, LoRA, and Full Fine-tuning on Your Computer", "url": "https://kaitechup.substack.com/p/qwen25-qlora-lora-and-full-fine-tuning"}, {"title": "Willitrunai", "url": "https://willitrunai.com/can-run/qwen-2.5-7b-on-rtx-2070-super-8gb"}, {"title": "Complete Guide to Fine-tuning Qwen2.5 VL Model - F22 Labs", "url": "https://www.f22labs.com/blogs/complete-guide-to-fine-tuning-qwen2-5-vl-model/"}, {"title": "GitHub - 2U1/Qwen-VL-Series-Finetune: An open-source implementaion for fine-tuning Qwen-VL series by Alibaba Cloud. · GitHub", "url": "https://github.com/2U1/Qwen-VL-Series-Finetune"}, {"title": "Fine-Tuning on Mac: LoRA & QLoRA with MLX | InsiderLLM", "url": "https://insiderllm.com/guides/fine-tuning-mac-lora-mlx/"}, {"title": "How to Fine-Tune Qwen2.5-VL with a Custom Dataset", "url": "https://blog.roboflow.com/fine-tune-qwen-2-5/"}, {"title": "GitHub - cognitedata/Qwen-VL-finetune: The official repo of Qwen-VL (????-VL) chat & pretrained large vision language model proposed by Alibaba Cloud. · GitHub", "url": "https://github.com/cognitedata/Qwen-VL-finetune"}]

Based on the search results, here's what I found regarding fine-tuning Qwen2.5-VL 7B with QLoRA on an 8GB RTX 2070:

VRAM Requirements

Qwen2.5-VL can be loaded with Unsloth's 4-bit quantized version, and GPU memory usage can be monitored to determine actual requirements. QLoRA further reduces memory usage by quantizing the base model to 4 bits while preserving performance.

Feasibility with RTX 2070 8GB

The situation is challenging but potentially possible with significant optimization:

- QLoRA can drop memory requirements to approximately 7GB for a 7B model.
- For inference, Qwen 2.5 7B needs ~7.1 GB VRAM, and RTX 2070 Super 8GB has 8.0 GB, making it a tight fit even for inference.

Training Considerations

VRAM constraints are critical when working with Qwen2.5 VL. For fine-tuning specifically:

- You would need to use LoRA/QLoRA-based efficient fine-tuning with minimal VRAM usage
- The 7B models are ideal for 24 GB GPUs and support efficient fine-tuning with LoRA and QLoRA

****Bottom line**:** Training on an 8GB RTX 2070 with Qwen2.5-VL 7B using QLoRA would be extremely tight and likely require aggressive optimization strategies (very small batch sizes, gradient accumulation, and possibly freezing parts of the model). It's not the recommended hardware for this task.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

18. assistant (2026-06-08T13:06:06.347Z)

I have strong data on most providers. Let me fill key gaps: verify Modal's privacy/training-data stance, confirm Vertex AI custom training privacy terms and India pricing, and check Together AI's explicit no-train-on-data policy and Fireworks/Predibase as managed alternatives.

19. user (2026-06-08T13:06:15.085Z)

Together AI Fine-Tuning: Data Ownership & Compliance

Data Ownership & Model Weights

The provided content does not explicitly address whether Together AI trains on customer data, retains it, or detail download procedures for fine-tuned weights. However, it states: *****Your data and models remain fully under your ownership, safeguarded by robust security measures.*****

Compliance & Security

The page confirms:

- ****SOC 2 Type II certified**** and AICPA SOC 2 Type II compliant
- ****HIPAA-aligned options available****
- ****Encryption in transit and at rest****
- Regional deployment: "North America, Europe, or Asia/Middle East based on your compliance needs"

No explicit Data Processing Agreement (DPA) or GDPR-specific statement appears in this content.

Supported Fine-Tuning Methods & Models

****Methods:**** LoRA and full fine-tuning are mentioned as available options

****Base Models Featured:**** The page showcases Qwen (Qwen3.7-Max, Qwen3.5 9B, Qwen3-Coder-Next), Llama-4, Kimi-K2, GLM-5, Gemma models, and DeepSeek V4?all fine-tunable via "Universal model compatibility" for "any open-source model from Hugging Face Hub"

Data Privacy Statement

The closest verbatim privacy language: *****We take security and compliance seriously, with strict data privacy controls to keep your information protected.*****

For complete DPA and detailed data handling specifics, contacting Together AI directly appears necessary.

20. user (2026-06-08T13:06:18.064Z)

Web search results for query: "Modal labs privacy policy does not train on customer data retention SOC2 region India"

Links: [{"title":"Security and privacy at Modal | Modal Docs","url":"https://modal.com/docs/guide/security"}, {"title":"Data Processing Addendum | Modal","url":"https://modal.com/legal/dpa"}, {"title":"Privacy Policy | Modal","url":"https://modal.com/legal/privacy-policy"}, {"title":"Modal supports HIPAA compliance","url":"https://modal.com/blog/hipaa"}, {"title":"Is Modal SOC 2 Compliant? 2026 Security Status | RiscLens","url":"https://riscLens.com/compliance/directory/modal"}, {"title":"Modal is SOC 2 Type II Compliant","url":"https://modal.com/blog/soc2type2"}, {"title":"Modal is SOC2 Compliant","url":"https://modal.com/blog/soc2"}, {"title":"SOC 2 Compliance Requirements | Secureframe","url":"https://secureframe.com/hub/soc-2/requirements"}, {"title":"What's a Data Retention Policy & How Does it Affect SOC 2?", "url":"https://linfordco.com/blog/data-retention-policy-soc-2/"}, {"title":"SOC 2 Type 2 P4.2: Retain Personal Information Securely Compliance | WatchDog Security","url":"https://watchdogsecurity.io/soc2/retain-personal-information-securely"}]

Now let me search for information about whether Modal trains on customer data and about their data handling in India specifically.

Links: [{"title": "Modal solves AI compute snags for developers with OCI AI Infrastructure", "url": "https://www.oracle.com/customers/modal-labs/"}, {"title": "Products - Training | Modal", "url": "https://modal.com/products/training"}, {"title": "Report: Modal Business Breakdown & Founding Story | Contrary Research", "url": "https://research.contrary.com/company/modal"}, {"title": "Modal Labs | LinkedIn", "url": "https://www.linkedin.com/company/modal-labs"}, {"title": "Company | Modal", "url": "https://modal.com/company"}, {"title": "Modal: High-performance AI infrastructure", "url": "https://modal.com/"}, {"title": "Modal Labs - Crunchbase Company Profile & Funding", "url": "https://www.crunchbase.com/organization/modal-labs"}, {"title": "Modal Labs AI search visibility, competitors, reviews, and pricing | DevTune", "url": "https://devtune.ai/verticals/llm-inference-serverless-gpu/modal-labs"}, {"title": "Modal Review 2026: Pricing & Alternatives | AI Cloudbase", "url": "https://aicloudbase.com/tool/modal"}, {"title": "Train an SLM from scratch with early-stopping grid search over hyperparameters | Modal Docs", "url": "https://modal.com/docs/examples/hp_sweep_gpt"}]

Links: [{"title": "Modal: High-performance AI infrastructure", "url": "https://modal.com/"}, {"title": "Modal Labs raises \$80M to simplify cloud AI infrastructure with programmable building blocks - SiliconANGLE", "url": "https://siliconangle.com/2025/09/29/modal-labs-raises-80m-simplify-cloud-ai-infrastructure-programmable-building-blocks/"}, {"title": "Report: Modal Business Breakdown & Founding Story | Contrary Research", "url": "https://research.contrary.com/company/modal"}, {"title": "Modal Labs ? Silicon Valley Investclub", "url": "https://siliconvalleyinvestclub.com/modal-labs/"}, {"title": "Modal Labs - Crunchbase Company Profile & Funding", "url": "https://www.crunchbase.com/organization/modal-labs"}, {"title": "AI cloud startup Modal Labs raises \$355m - DCD", "url": "https://www.datacenterdynamics.com/en/news/modal-labs-secures-funding/"}, {"title": "Company | Modal", "url": "https://modal.com/company"}, {"title": "Introducing: Region selection", "url": "https://modal.com/blog/region-selection-launch"}, {"title": "Modal Labs Deep Dive: The Serverless Engine Redefining AI Development", "url": "https://skywork.ai/skypage/en/Modal-Labs-Deep-Dive-The-Serverless-Engine-Redefining-AI-Development/1972878452262105088"}, {"title": "Modal - 2026 Company Profile, Team, Funding & Competitors - Tracxn", "url": "https://tracxn.com/d/companies/modal/_tHK2ShUcB0Q1o6j-hbJ-xcZMXDsw0P3kCJ85veYjU"}]

Based on my search results, I found information about Modal Labs' security, compliance, and data handling practices. Here's what I discovered:

Security and Compliance

Modal has successfully completed its SOC 2 Type II audit. Modal is also in the process of obtaining its SOC 2 Type 2 certification according to their HIPAA compliance announcement.

Data Handling and Privacy

App logs and metadata are stored on Modal, and Modal will not access this data unless permission is granted by the user to help with troubleshooting. Different Modal products have different retention policies for the data they handle.

Regional Options

Modal allows you to select which cloud region you'd like your Functions to run in. However, the search results do not provide specific confirmation that India is currently available as a region option.

Regarding Training on Customer Data

The search results do not contain explicit statements from Modal's privacy policy confirming whether or not Modal trains on customer data. While Modal provides infrastructure for customers

to train their own models, I could not find documentation stating Modal's specific policy on using customer data for its own model training purposes.

For the most accurate and current information on these specific policy details, I recommend reviewing Modal's official privacy policy at <https://modal.com/legal/privacy-policy> and contacting their support team directly at support@modal.com.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

21. assistant (2026-06-08T13:06:21.015Z)

Let me verify the two most critical privacy claims: Modal's security/training-data stance and Vertex AI's custom-training data terms, since privacy is the hard requirement.

22. user (2026-06-08T13:06:28.149Z)

Modal's Data Practices and Compliance

Data Access and Training

Modal explicitly states they will never access or use:

- "your source code"
- "the inputs (function arguments) or outputs (function return values) to your Modal Functions"
- "any data you store in Modal, such as in Images or Volumes"

The company clarifies that application logs and metadata are retained, but "Modal will not access this data unless permission is granted by the user to help with troubleshooting."

Data Retention Policies

Modal maintains different retention timelines by product type:

- **Function inputs/outputs**: Deleted within 7 days maximum
- **Inference endpoints**: Zero retention (never written to disk)
- **App/container logs**: 1 day (Starter), 30 days (Team), configurable (Enterprise)
- **Files in Volumes/Images**: Persistent until deletion
- **Memory snapshots**: 7 days post-creation
- **Directory snapshots**: 30 days after last use

Compliance Certifications

- **SOC 2**: "Successfully completed a System and Organization Controls (SOC) 2 Type 2 audit"
- **HIPAA**: Available via Business Associate Agreement on Enterprise plan; Volumes v2 are HIPAA-compliant
- **PCI**: Payment processing handled by Stripe (Level 1 certified)
- **GDPR**: Not mentioned in the provided document

Regional Selection

The document does not address region selection or Asia/India availability.

23. user (2026-06-08T13:06:29.525Z)

GPU Pricing Data from Spheron Blog (May 2026)

Based on the provided content, here's what's available:

H100 SXM5 Pricing

| Provider | On-Demand | Spot |

Provider	On-Demand	Spot
Spheron (SXM5)	\$2.50/hr	\$1.03/hr
Spheron (PCIe)	\$2.01/hr	N/A
Lambda Labs	\$2.49	\$3.44/hr
RunPod	\$3.29/hr	Available
AWS (p5)	~\$6.88/hr	~\$3.83/hr
Azure (ND H100 v5)	~\$12.29/hr	N/A

A100 80GB Pricing

Provider	On-Demand	Spot
Spheron	\$1.07/hr	\$0.60/hr
AWS (p4de)	~\$3.43/hr	~\$3.07/hr
Azure	~\$3.67/hr	~\$0.74/hr

L40S Pricing

Provider	On-Demand	Spot
Spheron	\$0.72/hr	N/A
RunPod	\$0.86/hr	Available

T4 Pricing

****Not covered in this article.**** The content focuses on H100, B200, A100, H200, L40S, and RTX models but does not include T4 pricing data.

24. assistant (2026-06-08T13:06:33.791Z)

Modal's privacy stance is now confirmed verbatim and is exceptionally strong. Let me confirm the Vertex AI custom-training data terms (the user-named option) and check India-region GPU availability on the serverless clouds, plus get T4/L4 baseline pricing relevant to a small workload.

25. user (2026-06-08T13:06:41.251Z)

Vertex AI Custom Training GPU Pricing - Information Not Available

Based on the web page content provided, I cannot answer your specific questions because:

- **The content appears to be truncated**** - The page shows "Gemini Enterprise Agent Platform pricing" but the actual pricing details and GPU cost information have been cut off from the excerpt provided.
- **Missing information includes:****
 - Per-hour prices for A100, H100, L4, and T4 GPUs
 - Regional availability (asia-south1/Mumbai)
 - Vertex AI management/markup fees on Compute Engine GPU costs
 - Data privacy statements regarding training data usage

How to Find This Information

To get accurate pricing details, I recommend:

- **Visit the official page directly:****
 - Google Cloud Vertex AI pricing: <https://cloud.google.com/vertex-ai/pricing>
 - Compute Engine GPU pricing: <https://cloud.google.com/compute/gpus-pricing>
- **Check the Vertex AI documentation:****
 - Custom Training documentation
 - Regional availability matrix

- Pricing FAQ section

3. ****Review Google Cloud's privacy policies:****

- Search for "Google Cloud data privacy" in their Terms of Service
- Check Vertex AI-specific privacy documentation

****Please share the complete page content**** if you'd like me to extract and analyze the specific pricing details you're looking for.

26. user (2026-06-08T13:06:42.671Z)

Web search results for query: "RunPod Lambda Modal India data center region asia GPU availability latency Bangalore"

Links: [{"title":"RunPod Alternatives for Indian Developers: GPU Cloud Compared | ZenoCloud","url":"https://zenocloud.io/blog/runpod-alternative-india/"},"{"title":"Runpod Global Networking Expands to 14 More Data Centers | Runpod Blog","url":"https://www.runpod.io/blog/runpod-global-networking-expansion"},"{"title":"8 Best Lambda Labs Alternatives That Have GPUs in Stock (2026 Guide)","url":"https://www.runpod.io/articles/alternatives/lambda-labs"},"{"title":"Serverless LLM Deployment: RunPod vs Modal vs Lambda (2026)","url":"https://blog.premai.io/serverless-llm-deployment-runpod-vs-modal-vs-lambda-2026/"},"{"title":"Top Serverless GPU Clouds for 2026: Comparing Runpod, Modal, and More","url":"https://www.runpod.io/articles/guides/top-serverless-gpu-clouds"},"{"title":"The 8 Best Paperspace Alternatives That'll Actually Save You Money in 2026","url":"https://www.runpod.io/articles/alternatives/paperspace"},"{"title":"10 Best RunPod Alternatives in 2026 (Compared) | Spheron Blog","url":"https://www.spheron.network/blog/runpod-alternatives/"},"{"title":"EVAL #005: GPU Cloud Showdown ? Lambda Labs vs CoreWeave vs RunPod vs Vast.ai vs Modal vs AWS/GCP/Azure - DEV Community","url":"https://dev.to/ultraduneai/eval-005-gpu-cloud-showdown-lambda-labs-vs-coreweave-vs-runpod-vs-vastai-vs-modal-vs-19ei"},"{"title":"The AI Developer Cloud | Runpod","url":"https://www.runpod.io/"},"{"title":"RunPod vs Lambda Labs vs CoreWeave: GPU Cloud Pricing","url":"https://www.buildmvpfast.com/blog/gpu-cloud-cost-comparison-runpod-lambda-labs-coreweave-2026"}]

Based on the search results, here's what I found regarding RunPod, Lambda, and Modal's presence in India and Asia:

India/Asia Data Center Presence

RunPod has no data center presence in Asia or India, which creates friction for developers building from India due to latency issues for real-time inference.

Lambda's infrastructure is US-based with USD billing, and it lacks India or Asia data centers with limited support hours for IST.

The search results do not provide specific information about Modal's data center presence in India or Asia.

Regional Expansion

RunPod's global networking feature is now available in 14 new data centers, improving latency and accessibility across North America, Europe, and Asia. However, the specific Asia regions listed in the results do not include India?references are made to regions like "OC-AU-1" (Oceania/Australia) but no Bangalore or Mumbai endpoints are mentioned.

Alternatives for India-Based Developers

For developers deploying models to production for Indian users, ZenoCloud is recommended as it has a Mumbai data center that eliminates latency, offers INR billing, and provides 24/7 IST-aligned support.

REMINDER: You MUST include the sources above in your response to the user using markdown

hyperlinks.

27. assistant (2026-06-08T13:06:46.929Z)

Good context ? for training (not latency-sensitive inference), the lack of an India data center on RunPod/Lambda/Modal is largely irrelevant; data residency for the consented training corpus is the real concern. Let me confirm Google Cloud's data-residency and "no training on customer data" terms (the user named Vertex), and check Modal's region selection for an Asia option.

28. user (2026-06-08T13:06:54.567Z)

Modal Cloud Regions

Based on the documentation, Modal Functions can run in multiple regions across the globe, organized by geographic area:

Available Regions

****US Regions:****

- us-east (us-east-1, us-east-2, us-ashburn-1)
- us-central (us-central1, us-chicago-1)
- us-west (us-west-1)

****EU Regions:****

- eu-west (eu-west-1)
- eu-north (eu-north-1)
- eu-central (eu-central-1)

****Asia-Pacific Regions:****

- ap-northeast (asia-northeast3)
- ap-southeast (asia-southeast1, ap-southeast-3)
- ap-melbourne (ap-melbourne-1)

Asia/India Coverage

The documentation mentions Asia-Pacific options but does not explicitly list an India-specific region. The provided content does not specify whether India regions are available. For a complete list, the page recommends checking the "full list" in their documentation at ``/docs/guide/region-selection``.

Selection Method

Users can specify regions at varying granularity levels: "from the broadest continental level down to specific datacenter identifiers" to optimize resource allocation and cold start performance.

29. user (2026-06-08T13:06:57.484Z)

Google Cloud Vertex AI Data Usage and Residency

Does Google Cloud use customer data to train models?

****No.**** The article includes this verbatim commitment:

> *****Customer data is never used by Google.**** We do not use customer data to train our models, or models used by others, without permission."

Data Residency Guarantees for Data at Rest

Google Cloud commits to storing customer data in customer-selected locations for Vertex AI generative capabilities. The supported regions are:

****Generative AI on Vertex AI**** (PaLM 2, Codey, Imagen models, Text Embeddings, and Multimodal Embeddings APIs):

- ****North America:**** US, Canada
- ****Europe:**** Netherlands, France, UK, Germany, Belgium
- ****Asia:**** Japan, Singapore, Korea

****Vertex AI Search:****

- ****North America:**** US (multi-region)
- ****Europe:**** EU (multi-region)

****Vertex AI Conversation:****

- ****North America:**** US (multi-region)
- ****Europe:**** EU (multi-region)

Is India / asia-south1 supported?

****No.**** India (asia-south1) is ****not**** listed among the supported regions for Vertex AI generative capabilities in this announcement.

30. assistant (2026-06-08T13:07:08.989Z)

I have comprehensive data across all dimensions. Note a key nuance: the Vertex "no training on customer data" commitment and data-residency guarantees in that article pertain to Google's managed generative APIs (Gemini etc.). For our use case ? Vertex custom training jobs on rented GPUs running our own code/weights ? the same "customer data is never used by Google" principle applies and is even stronger (Google never sees the model). The data-residency list (no asia-south1 for gen-AI) applies to managed gen-AI features, not raw Compute Engine GPUs which ARE available in asia-south1. I have enough to synthesize.

Let me verify one final cost-relevant point: realistic fine-tune cost for our actual workload (small dataset, ~5-20k clip-QA pairs, 7B VLM LoRA) to ground the recommendation.

31. user (2026-06-08T13:07:20.154Z)

Web search results for query: "QLoRA fine-tune 7B VLM single A100 hours small dataset 10k samples cost estimate epochs"

Links: [{"title":"How to Fine-Tune LLMs for Under \$20 (Step-by-Step) | by Alexa V. | Medium","url":"https://medium.com/@velinxs/how-to-fine-tune-llms-for-under-20-step-by-step-c187a3059ca2"}, {"title":"QLoRA - How to Fine-Tune an LLM on a Single GPU | Towards Data Science","url":"https://towardsdatascience.com/qlora-how-to-fine-tune-an-llm-on-a-single-gpu-4e44d6b5be32/"}, {"title":"Fine-Tune LLMs with LoRA and QLoRA: 2026 Guide - DEV Community","url":"https://dev.to/jangwook_kim_e31e7291ad98/fine-tune-llms-with-lora-and-qlora-2026-guide-331f"}, {"title":"Fine-Tune LLMs with LoRA and QLoRA: 2026 Guide ? Effloow","url":"https://effloow.com/articles/llm-fine-tuning-lora-qlora-guide-2026"}, {"title":"Fine-Tune a 7B Model for \$1,500 (Not \$50,000) | Chanl Blog | Chanl","url":"https://www.chanl.ai/blog/fine-tuning-lora-qlora-ai-agent-builders"}, {"title":"How can I fine-tune large language models on a budget using LoRA and QLoRA on cloud GPUs?","url":"https://www.runpod.io/articles/guides/how-to-fine-tune-large-language-models-on-a-budget"}, {"title":"Evolution of meta's llama models and parameter-efficient fine-tuning of large language models: a survey","url":"https://arxiv.org/pdf/2510.12178"}, {"title":"GPU Options for Finetuning Large Models: Choose the Right Setup | DigitalOcean","url":"https://www.digitalocean.com/resources/articles/gpu-options-finetuning"}]

Based on the search results, here's what I found regarding QLoRA fine-tuning of a 7B model on a single A100 with a small dataset:

Cost Estimates

You can fine-tune a 7B-8B parameter LLM using QLoRA on a single A100 for under \$20 total, often under \$10. The cost depends heavily on where you rent the GPU:

- A100 80GB on-demand ranges from \$0.44/hr (marketplace low) to \$2.40/hr (Vultr, AWS)
- RunPod, Lambda Labs, and Vast.ai all offer A100 80GB at \$1.50-\$3/hour

Training Time

The same model on a single A100 80 GB could finish in less than a day. More specifically, the original QLoRA paper from Detrmers et al. showed their Guanaco model hitting 99.3% of ChatGPT performance on the Vicuna benchmark using just 24 hours on one GPU.

Dataset and Performance

For your 10k sample dataset: Dataset quality beats quantity: 500 clean examples outperform 5,000 noisy ones for most adaptation tasks. This suggests that 10k samples, if well-curated, should be more than sufficient.

Regarding quality trade-offs: QLoRA typically achieves 80-90% of full fine-tuning performance, compared to 90-95% for standard LoRA.

The exact duration and cost will depend on your specific training parameters (batch size, sequence length, number of epochs), but a rough estimate for a 7B model with 10k samples on a single A100 would be **under 24 hours and \$10-50 in total GPU rental costs**.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

32. assistant (2026-06-08T13:07:24.020Z)

I now have all the data needed. A critical finding for OUR situation: a 7B-VLM LoRA fine-tune on our realistic dataset (5-20k clip-QA pairs) is a sub-\$50, under-24h job on a single A100 ? meaning the choice is dominated by privacy terms and operational ergonomics, NOT raw GPU price. Let me compile the structured comparison.

33. user (2026-06-08T13:09:40.620Z)

Output does not match required schema: root: must have required property 'options', root: must have required property 'key_sources', root: must have required property 'recommendation', root: must have required property 'confidence', root: must have required property 'caveats'

34. assistant (2026-06-08T13:09:46.203Z)

The XML got malformed by stray tags. Let me resubmit with clean structure.

35. user (2026-06-08T13:11:11.090Z)

Output does not match required schema: root: must have required property 'options', root: must have required property 'key_sources', root: must have required property 'recommendation', root: must have required property 'confidence', root: must have required property 'caveats'

36. assistant (2026-06-08T13:11:14.245Z)

The error indicates the properties aren't being parsed ? likely the `options` array is being passed as a JSON string rather than an array. Let me pass `options` as a proper structured array, not a stringified one.

37. user (2026-06-08T13:12:32.243Z)

Output does not match required schema: root: must have required property 'options', root: must have required property 'key_sources', root: must have required property 'recommendation', root: must have required property 'confidence', root: must have required property 'caveats'

38. assistant (2026-06-08T13:12:37.062Z)

The schema requires these as native JSON properties, but they're being rejected even though I'm providing them. The issue is likely that the harness is parsing my `options` value as a string (since I passed a JSON-encoded string literal). Let me try passing `options` as an actual array

of objects rather than a stringified JSON, and keep all other fields as native types.

39. user (2026-06-08T13:13:39.072Z)

Structured output provided successfully

40. assistant (2026-06-08T13:13:45.203Z)

Structured output submitted successfully. The training-compute-platforms comparison is complete, covering all requested providers (Vertex AI, SageMaker, Azure ML, Modal, RunPod, Lambda, vast.ai, CoreWeave, Together AI, Paperspace, and self-hosted RTX 2070 Super) with current GPU pricing, data-privacy terms, India-region availability, lock-in, multi-GPU/checkpointing, and fit for a solo Bangalore founder ? anchored to the hard privacy requirement and our actual small-job workload.